

Source-Backed Local-to-Global Gluing for Formal H1 Occurrence Crosswalks

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Abstract

The missing H1 occurrence crosswalk need not be discovered as one monolithic table. This paper proves the finite descent theorem needed to construct it from local patches. Let a finite family $\{U_i\}$ cover the archived cuts. Suppose each $c \in U_i$ has a finite nonempty local formal row family, and suppose exact typed, weight-preserving bijections identify the row families on pairwise overlaps. If the bijections satisfy the identity, inverse, and triple-overlap cocycle laws, the quotient of the disjoint union is a global formal row family. All preserved payloads descend uniquely; the result is unique up to the unique typed isomorphism commuting with the local maps. If each local column has exact rational weight one, then every global column also has exact weight one.

This theorem is formal and conditional. A synthetic two-patch fixture glues four local row copies to three global rows and verifies exact conservation, while a three-patch regression tests the cocycle nonvacuously. Neither fixture has production semantics. TPC-163 finds zero theorem-backed production actual-occurrence edges in the frozen declared corpus, so no production local patch system is currently available.

Most importantly, the paper separates three gates: formal occurrence totality, actual active support, and canonical/minimal representation. Local-to-global gluing can close only the first. It cannot promote a formal row to nonzero physical support or choose a canonical actual parent.

Keywords: fixed shift; finite descent; gluing; occurrence provenance; exact column conservation; proof interface.

Claim boundary. The theorem says what follows *if* source-backed compatible local patches are supplied. The current production corpus supplies none. Synthetic fixtures have no theorem semantics. We prove no actual active support, canonical/minimal actual representation, production H1 crosswalk, positive fixed- X L2, prime-pair lower bound, or twin-prime theorem.

1 From a monolithic blocker to a descent problem

TPC-163 gives a source-locator contract and finds no qualifying production actual-occurrence edge in its frozen corpus [2]. TPC-164 nevertheless solves the archive-address problem: the unique minimum separating key in its eight-field dictionary is

$$\kappa(c) = (\ell(c), k(c), d_{\text{nat}}(c), j_L(c), j_K(c)). \quad (1.1)$$

This key does not create actual occurrences, but it lets independent local constructions refer to the same source cut unambiguously [3].

The next structural question is therefore:

can compatible local row families be assembled without losing types or exact coefficients?

The answer is yes, by an elementary finite quotient theorem. This does not solve the source problem; it isolates exactly which local data would suffice.

2 Typed local patch system

Definition 2.1 (Local formal row system). Let $\mathcal{C}$ be a finite set of archived cut addresses, and let $\{U_i\}_{i \in I}$ be a finite cover. For every $c \in U_i$, let $\mathcal{R}_i(c)$ be a finite nonempty set of local formal rows. A row $r \in \mathcal{R}_i(c)$ carries:

- (i) an archived source address $\kappa(c)$;
- (ii) a typed payload $T_i(r)$;
- (iii) an exact weight $w_i(r) \in \mathbb{Q}$; and
- (iv) , in production use, a source record satisfying the path-hash-theorem-formula-derivation contract of TPC-163.

Local column conservation means

$$\sum_{r \in \mathcal{R}_i(c)} w_i(r) = 1 \quad (c \in U_i). \quad (2.1)$$

The typed payload may contain a formal occurrence symbol, stage, parent, multiplier, and downstream target. The theorem treats those as data to be preserved; it does not assert that they are true of an actual carrier.

Definition 2.2 (Descent datum). For every $c \in U_i \cap U_j$, an overlap map is a bijection

$$\phi_{ji,c} : \mathcal{R}_i(c) \longrightarrow \mathcal{R}_j(c) \quad (2.2)$$

such that

$$T_j(\phi_{ji,c}r) = T_i(r), \quad w_j(\phi_{ji,c}r) = w_i(r). \quad (2.3)$$

The maps form a *descent datum* if

$$\phi_{ii,c} = \text{id}, \quad \phi_{ij,c} = \phi_{ji,c}^{-1}, \quad \phi_{ki,c} = \phi_{kj,c} \circ \phi_{ji,c} \quad (2.4)$$

on every applicable pairwise or triple overlap.

3 Local-to-global gluing theorem

Theorem 3.1 (Finite typed gluing). Let $(\mathcal{R}_i, T_i, w_i, \phi_{ji})$ be a local formal row system with descent datum. Then:

- (a) there is a global finite nonempty row family $\mathcal{R}(c)$ for every $c \in \mathcal{C}$;
- (b) each local family has a canonical bijection $\iota_{i,c} : \mathcal{R}_i(c) \rightarrow \mathcal{R}(c)$, and $\iota_{j,c} \phi_{ji,c} = \iota_{i,c}$;
- (c) the payload and weight descend uniquely to functions $T : \mathcal{R}(c) \rightarrow \mathcal{T}$ and $w : \mathcal{R}(c) \rightarrow \mathbb{Q}$;
- (d) the global object is unique up to the unique typed, weight-preserving isomorphism commuting with all $\iota_{i,c}$; and
- (e) if (2.1) holds, then

$$\sum_{r \in \mathcal{R}(c)} w(r) = 1 \quad (c \in \mathcal{C}). \quad (3.1)$$

Proof. Fix $c \in \mathcal{C}$. Form the disjoint union

$$D(c) = \coprod_{i:c \in U_i} \mathcal{R}_i(c).$$

Declare $(i, r) \sim (j, s)$ if $s = \phi_{ji,c}(r)$. Identity and inverse give reflexivity and symmetry, and the cocycle gives transitivity. Thus $\sim$ is an equivalence relation. Put

$$\mathcal{R}(c) = D(c)/\sim, \quad \iota_{i,c}(r) = [i, r]. \quad (3.2)$$

For fixed i, j containing c , the overlap bijection identifies each row of $\mathcal{R}_i(c)$ with exactly one row of $\mathcal{R}_j(c)$. Consequently every equivalence class has exactly one representative in each participating patch, and each $\iota_{i,c}$ is bijective. Equation (2.3) makes

$$T([i, r]) = T_i(r), \quad w([i, r]) = w_i(r)$$

well-defined. Uniqueness of these descended functions is immediate because the local maps are jointly surjective.

If another global family $\mathcal{R}'(c)$ has compatible local bijections $\iota'_{i,c}$, define

$$\Psi_c([i, r]) = \iota'_{i,c}(r).$$

Compatibility makes Ψ_c well-defined. It is the only map commuting with the local maps, and the same construction in reverse gives its inverse. This proves the stated unique isomorphism.

Finally choose any i with $c \in U_i$. Since $\iota_{i,c}$ is a weight-preserving bijection,

$$\sum_{r \in \mathcal{R}(c)} w(r) = \sum_{r \in \mathcal{R}_i(c)} w_i(r) = 1,$$

which proves (3.1). □

Corollary 3.2 (Matrix conservation). *Let L_i send the basis vector e_c to the exact weighted sum of local rows in $\mathcal{R}_i(c)$. Under the hypotheses of [theorem 3.1](#), the L_i descend to a global matrix L , and*

$$\mathbf{1}^T L = \mathbf{1}^T. \tag{3.3}$$

Proof. The overlap maps identify equal typed coefficients, so the columns descend. Equation (3.1) is exactly (3.3) column by column. □

4 Why the cocycle is indispensable

Pairwise compatibility alone does not force consistent identifications around a triple overlap. If $\phi_{31,c} \neq \phi_{32,c}\phi_{21,c}$, a row in patch 1 gets two competing representatives in patch 3. The binary relation generated by pairwise maps still has a quotient, but a chosen local map need no longer be injective into it; the asserted common row family can collapse. Thus the cocycle is exactly the condition used in the injectivity step of [theorem 3.1](#).

The audit tests this nonvacuously. A three-patch, two-row fibre uses identity maps on all overlaps. Swapping only the direct map from the first patch to the third preserves exact payloads, weights, and pairwise bijectivity, but violates the triangle equation. The validator rejects it.

5 Synthetic nonvacuity certificate

The public sample uses

$$U_1 = \{c_1, c_2\}, \quad U_2 = \{c_2, c_3\}.$$

Each local cut has one row of rational weight 1. The two copies over c_2 have the same typed payload and are identified by the overlap bijection. Therefore:

quantity	value
cut count	3
patch count	2
local row copies	4
global formal rows	3
every local/global column sum	1.

This demonstrates that the serialization and quotient algorithm are reachable. Every record is explicitly marked `SYNTHETIC_REACHABILITY` with theorem semantics false.

6 Three gates that gluing must not merge

Definition 6.1 (Separated production gates). For a proposed production carrier, define:

- (G1) **formal occurrence totality**: every archived cut address has at least one globally glued formal row;
- (G2) **actual active support**: every required row is an actual carrier occurrence with the required nonzero physical coefficient; and
- (G3) **canonical/minimal representation**: an independent theorem selects or characterizes its canonical/minimal actual parent representation.

Proposition 6.2 (No gate promotion). *Theorem 3.1 can establish G1 from its local hypotheses. It does not imply G2 or G3.*

Proof. The construction in (3.2) uses only finite sets, typed payload equality, bijections, and exact formal weights. It contains no predicate that embeds a row into the actual arithmetic carrier or proves its physical coefficient nonzero, so it cannot imply G2. It also imposes no equivalence or optimization theorem on actual parents, so it cannot imply G3. Synthetic systems satisfying all gluing hypotheses while declaring both actual predicates undecided give explicit models of the separation. $\square$

TPC-155’s witness verifier [1] checks a declared interface. Even a successful format and conservation check would not, by itself, make an unsourced active-support or canonicity statement true. Those fields require their own source-backed theorems.

7 Current production evaluation

The imported source locks give:

Production object	Status
source-backed local actual-occurrence edges	0 in frozen corpus
local patch family	NOT-TESTABLE
overlap bijections and cocycle	NOT-TESTABLE
formal occurrence totality	NOT-TESTABLE
actual active support	NOT-TESTABLE
canonical/minimal representation	NOT-TESTABLE

The formal gluing theorem is proved, but its production hypotheses are not currently instantiated. This is not a negative existence theorem. New source-backed local occurrence results could populate the interface.

Reproducible audit. The generator source-locks TPC-163/164, recomputes the quotient and exact rational sums, and rejects real mutations of overlap, payload, weight, cocycle, evidence mode, production edges, and gate status.

8 Conclusion

Compatible source-backed local patches can build a global formal H1 crosswalk with exact conservation. Producing the local actual edges, proving actual active support, and proving canonical/minimal representation remain three distinct tasks that the next decision must preserve as parallel roots.

References

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